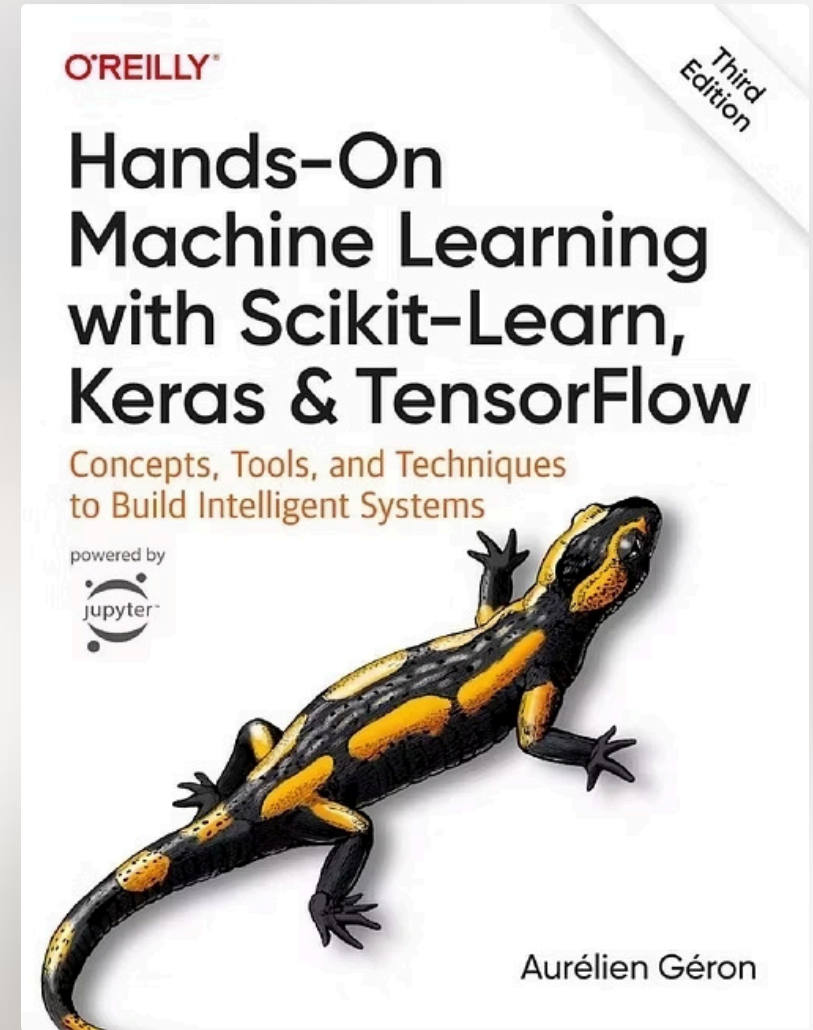


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 08/07/2025



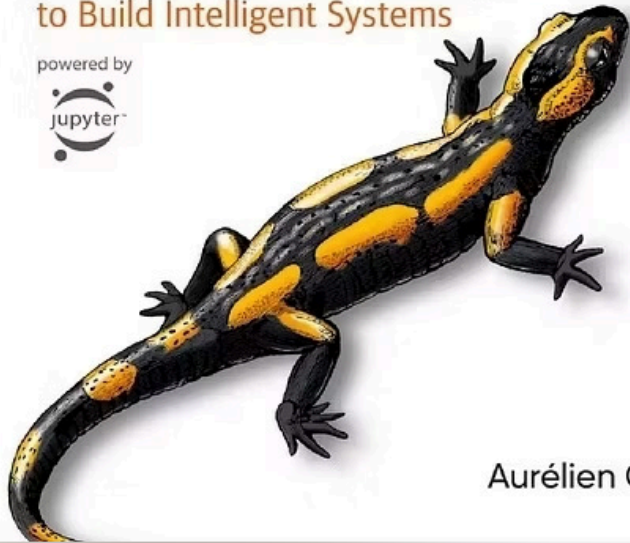
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Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
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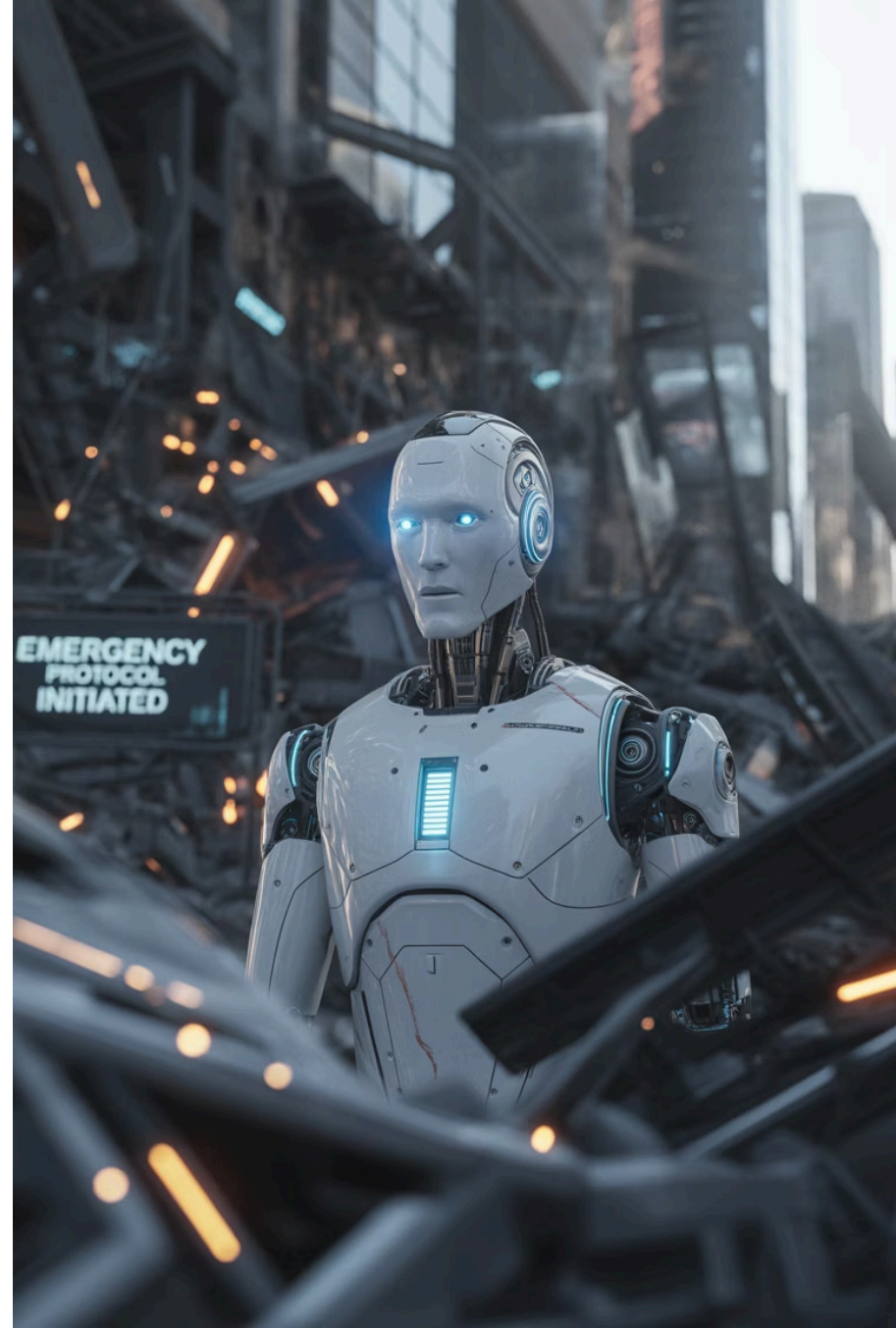
Aurélien Géron

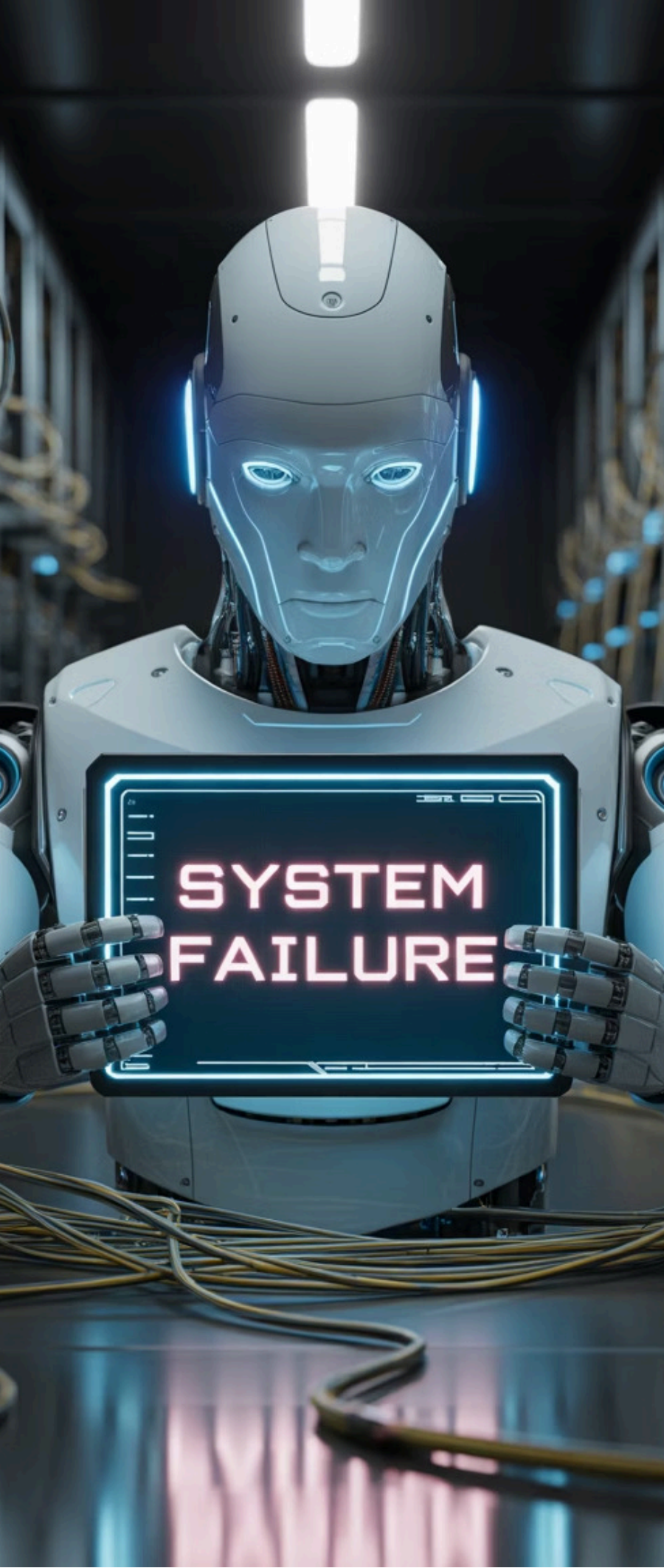
Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. **Main Challenges of Machine Learning**
6. Testing and Validating





Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - **Poor-Quality Data**
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back: Reviewing the Chapter So Far
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch

Main Challenges of Machine Learning

Insufficient Quantity of Training Data, Non-representative Training Data, **Poor-Quality Data**, Irrelevant Features, Overfitting the Training Data, Underfitting the Training Data, Stepping Back: Reviewing the Chapter So Far



Poor-Quality Data: Overview

Impact on ML

- **Errors, outliers, and noise in training data can make it harder for the system to detect underlying patterns.**
- **This leads to poor model performance.**

Data Cleaning

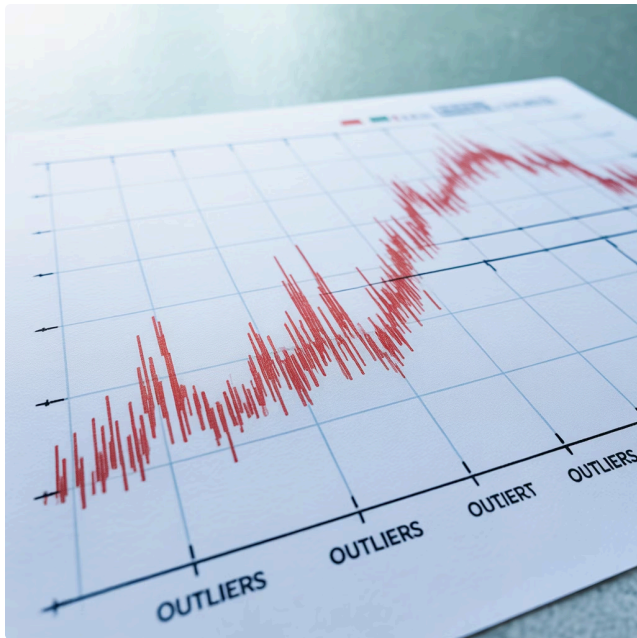
- **Cleaning up training data is often time-consuming but essential for improving model accuracy.**
- **Data scientists spend a significant amount of time cleaning data to ensure the quality of the dataset.**

Examples of Poor-Quality Data

Outliers

If some data instances are **clearly outliers**, it might be beneficial to:

- **Discard** them or attempt to fix the errors manually.



Missing Features

If some instances are missing a **few features** (e.g., 5% of customers did not specify their age), you need to decide how to handle it:

- **Ignore** the attribute entirely.
- **Ignore** the instances with missing values.
- **Fill in** the missing values (e.g., using the **median age**).
- Train **multiple models**: one with the **feature** and one **without** it.

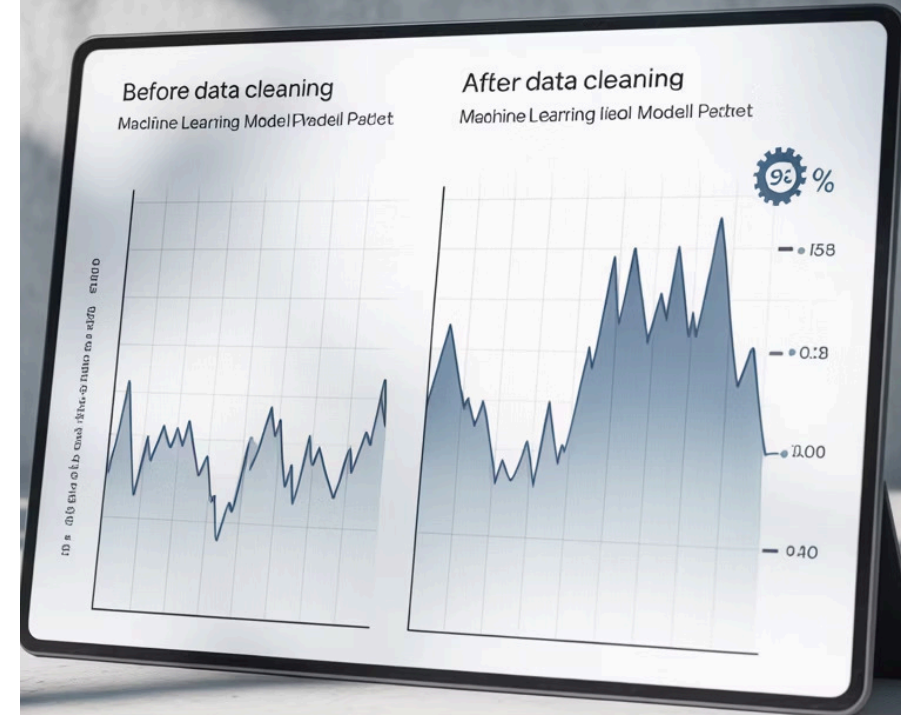
Key Takeaways

Importance of Cleaning Data:

Cleaning poor-quality data is essential to ensure that the **model** can **identify** the correct **patterns** and **generalize** well.

Handling Errors and Missing Data:

Decisions about how to deal with **outliers** and **missing data** significantly affect the **quality** of the **model** and its **predictions**.





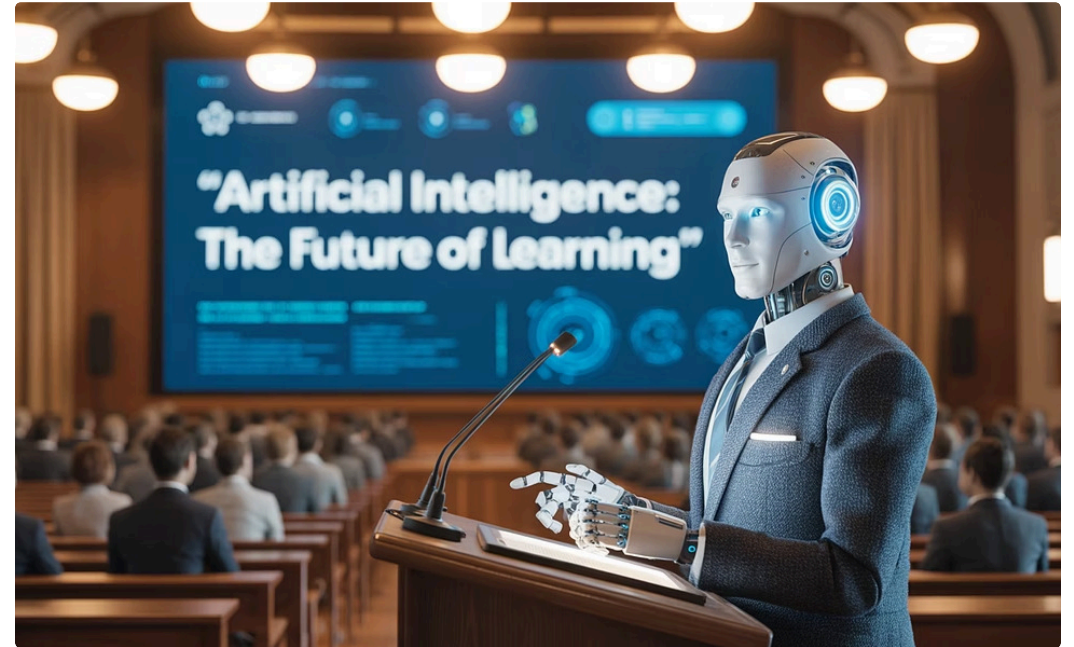
Next: Irrelevant Features

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts